MSIN0094 Third Assignment

Due Friday 10am, Dec 13, 2024

Candidate number: TBJY6

Word count: 1997

1. Descriptive Analytics (20 pts)

Q1 From `data_full`, generate a new variable, `final_price`, which is the actual retail price for each week (i.e., Recommended Retail Price after discounts). (**8pts**)

- Write your code below to generate final_price from RRP and discount. (**2pts**)

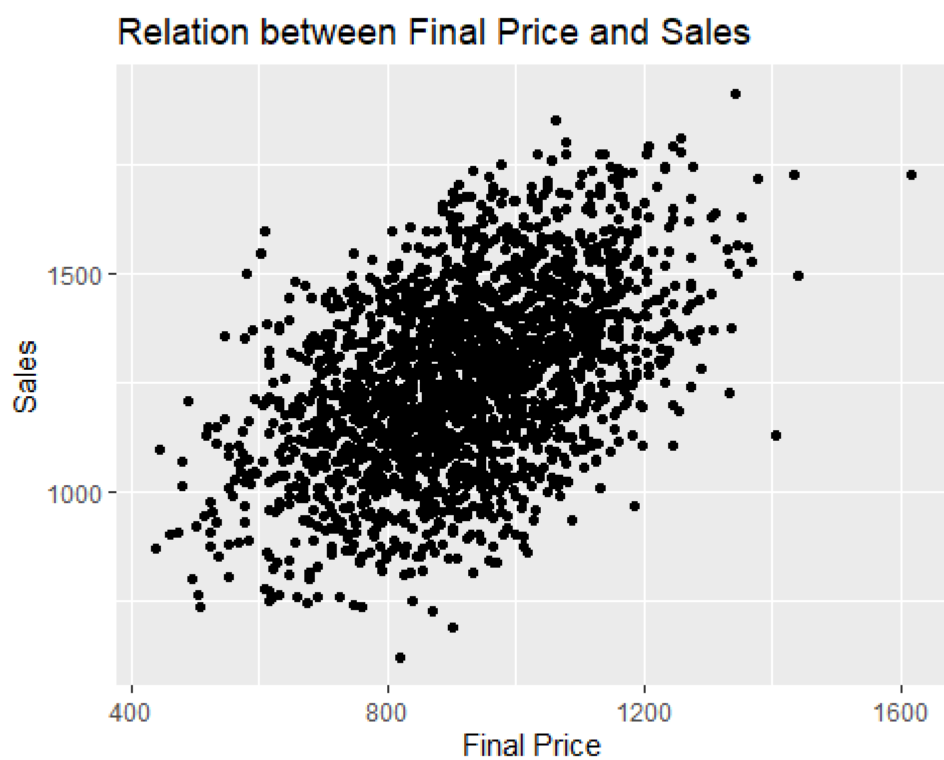
```
# write your codes below
data_full <- data_full %>%
  mutate(final_price = RRP * (1-discount))

glimpse(data_full)

Rows: 2,080
Columns: 13
$ product_id       <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
1…
$ brand            <chr> "Samsung", "Samsung", "Samsung", "Samsung",
"Samsung…
$ technology       <chr> "OLED", "OLED", "OLED", "QLED", "OLED", "QLED",
"QLE…
$ resolution       <chr> "1080p", "4k", "4k", "1080p", "4k", "4k", "4k",
"4k"…
$ support_HDR      <int> 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0,
1…
$ screensize       <chr> "50-59", "30-39", "30-39", "60+", "50-59", "40-49",
…
$ week_id          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1…
$ sales            <int> 1745, 1444, 1183, 1474, 1647, 1461, 1626, 1530,
1706…
$ RRP              <int> 1399, 1099, 1049, 949, 1299, 999, 1249, 1349, 1349,
…
$ discount         <dbl> 0.12, 0.12, 0.18, 0.13, 0.17, 0.12, 0.07, 0.15,
0.14…
$ marketing_expense <dbl> 3794.739, 3794.739, 3794.739, 3794.739, 3794.739,
37…
$ cost_shifter     <dbl> 618.6324, 571.6035, 644.6133, 593.3050, 646.6439,
60…
$ final_price      <dbl> 1231.12, 967.12, 860.18, 825.63, 1078.17, 879.12,
11…
```

- Visualise using scatter plot the relationship between final price and sales. Tips: you can use ggplot2 and geom_point to create the scatter plot. Write your code below to create the scatter plot. (**2pts**)

```
# write your codes below
library(ggplot2)
ggplot(data_full, aes(x=final_price, y=sales)) +
  geom_point() +
  labs(
    title="Relation between Final Price and Sales",
    x="Final Price",
    y ="Sales"
  )
```



- Do you observe a positive or negative relationship between final price and sales? Is this relationship causal? Why or why not? (**4pts**, 150 words)

Examining the scatterplot, it is clear that the relationship between final price and sales is positive. As the final price increases, sales also tend to rise. While this may appear counterintuitive from a traditional demand perspective, it is likely explained by product characteristics. Higher-priced TVs are typically larger or higher-quality models (e.g., OLED or 60-inch screens), which naturally attract more customers and therefore sell more units. In this dataset, price functions as a proxy for product quality rather than simply being a cost to the consumer.

Nevertheless, this relationship should not be interpreted as causal. We cannot conclude that raising prices leads to higher sales. Numerous other factors—such as brand reputation, screen size, technology type, and marketing investment—also influence sales and are likely correlated with price. Without controlling for these confounding variables, the observed pattern reflects correlation rather than a causal effect of price.

Q2. Use `dplyr` to compute the average weekly dollar sales (final price * unit sales) for each brand across all weeks (i.e., the result should be 1 average per brand). Rank the brands from the highest average dollar sales to the lowest average dollar sales. (**6pts**) Which brand has the highest average weekly dollar sales? (**2pts**).

```
# write you code below
data_sales_by_brand <- data_full %>%
  mutate(dollar_sales = final_price * sales) %>%
  group_by(brand) %>% #group by func
  summarise(avg_weekly_dollar_sales = mean(dollar_sales, na.rm = TRUE)) %>%
  arrange(desc(avg_weekly_dollar_sales))

#summarsise(avg_weekly_dollar_sales = mean(dollar_sales, na.rm = TRUE)) %>%
#calc avg weekly dollar sales for each brand by taking the mean of
dollar_sales

#na.rm = True, ensure missing values are ignored

#arrange(desc(avg_weekly_dollar_sales))
#sorts brands from highest to lowest based on avg weekly dollar sales

glimpse(data_full)

Rows: 2,080
Columns: 13
$ product_id       <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
1…
$ brand            <chr> "Samsung", "Samsung", "Samsung", "Samsung",
"Samsung…
$ technology       <chr> "OLED", "OLED", "OLED", "QLED", "OLED", "QLED",
"QLE…
$ resolution       <chr> "1080p", "4k", "4k", "1080p", "4k", "4k", "4k",
"4k"…
$ support_HDR      <int> 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0,
1…
$ screensize       <chr> "50-59", "30-39", "30-39", "60+", "50-59", "40-49",
…
$ week_id          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1…
$ sales            <int> 1745, 1444, 1183, 1474, 1647, 1461, 1626, 1530,
1706…
$ RRP              <int> 1399, 1099, 1049, 949, 1299, 999, 1249, 1349, 1349,
```

```
…
$ discount          <dbl> 0.12, 0.12, 0.18, 0.13, 0.17, 0.12, 0.07, 0.15,
0.14…
$ marketing_expense <dbl> 3794.739, 3794.739, 3794.739, 3794.739, 3794.739,
37…
$ cost_shifter      <dbl> 618.6324, 571.6035, 644.6133, 593.3050, 646.6439,
60…
$ final_price       <dbl> 1231.12, 967.12, 860.18, 825.63, 1078.17, 879.12,
11…

# please do not modify.
# print out the ranking of brands based on average weekly dollar sales
data_sales_by_brand

# A tibble: 4 × 2
  brand   avg_weekly_dollar_sales
  <chr>                     <dbl>
1 Samsung                1277231.
2 Sony                   1226644.
3 LG                     1119492.
4 Philips                1086011.
```

Samsung has the highest average weekly dollar sales (1277231).

> ***Q3.*** In Marketing, we refer to brand equity as the additional sales a brand can obtain when everything else is equal, i.e., the causal effect of brands on sales. Does the above average sales ranking causally identify which brand has the highest brand equity? Why or why not? (**4pts**; 150 words)

The ranking of average weekly dollar sales does not allow us to casually identify which brand has the highest brand equity. While Samsung shows the highest average weekly dollar sales, this does not necessarily mean that Samsung has the highest brand equity, because the observed sales differences may be driven by multiple confounding factors that are not controlled for. For example, Samsung may offer a larger proportion of high-end and high priced OLED or 60 inch TVs, which naturally generate higher revenue regardless of brand strength. Differences in screen size, technology, pricing strategy, product features, and marketing expenditure all influence sales and are corelated with brand.

To isolate brand equity, we would need to compare brands holding all other products attributes constant, for instance, through a regression model that controls for price, screen size, technology type, HDR support, and marketing spend. Without such controls, the ranking reflects correlation, not causal brand equity.

2. Marketing Mix Modeling and Endogeneity (28pts)

> ***Q4.*** Run a Marketing Mix Modeling linear regression as follows (**6pts**):

- Run the linear regression below using `fixest` package (Equation 1 hereinafter) (**2pts**).

```r
# write you codes for the regression below

#sales = a + b * final_price + c*marketing_expense + ϵ
pacman::p_load(modelsummary, fixest)

ols_1 <- feols(
  sales ~ final_price + marketing_expense,
  data = data_full

)

# do not modify the code below; this is to print out the results

modelsummary(ols_1,
             stars = T,
             gof_map = c('nobs','r.squared'))
```

(Intercept)	421.271***
	(20.126)
final_price	0.618***
	(0.020)
marketing_expense	0.094***
	(0.003)
Num.Obs.	2080
R2	0.503

- p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
- Interpret the coefficients of `final_price`, including coefficients and statistical significance (**4pts**).

The coefficient of 0.618 for **final_price** indicates that, after controlling for marketing_expense, a one-unit increase in the TV's final price is linked to a predicted rise of 0.618 units in sales (quantity sold). The three asterisks (***) show that this effect is statistically significant at the 0.1% level ($p < 0.001$). In a multivariate regression, this coefficient reflects the isolated impact of the variable on the outcome while keeping all other included variables constant. Still, as highlighted in the descriptive analysis, this positive relationship is unexpected and likely represents correlation rather than true causation, potentially driven by unobserved influences.

Q5. Based on the regression coefficients reported above, discuss the endogeneity issues with `final_price` in Equation 1. For each endogeneity cause, explain the general definitions and then give concrete examples in Amazon's context. (**12pts**)

- General definition of each endogeneity cause (**6pts**, 200 words)

Endogeneity is an econometric issue that arises when an explanatory variable (such as final_price) is correlated with the error team, violating the assumption $E[\epsilon \mid X] = 0$. When endogeneity exists, the OLS regression coefficient [β] does not represent the true causal effect.

- In Amazon's context, concrete examples to illustrate each endogeneity cause (**6pts**, 200 words)

- **Omitted Variable Bias (OVB):** The positive OLS coefficient (0.618) implies that OVB likely exists because higher-priced TVs tend to be premium, high-specification models (e.g., larger panels, OLED screens, HDR capabilities, or well-known brands like Samsung). These omitted product attributes influence both the final_price and the number of units sold, yet they were excluded from Equation 1. As a result, the estimated price coefficient unintentionally reflects the impact of these unobserved quality characteristics, making the effect appear positive even though price itself may not increase sales.

- **Reverse Causality (Simultaneity):** Amazon relies on algorithmic pricing, meaning the firm continually adjusts discounts—and ultimately the final_price—based on expected demand signals such as seasonal trends, competitor movements, or inventory conditions. When Amazon forecasts higher sales in upcoming weeks, the algorithm may increase or decrease discounts to either capture additional volume or maximize margin. This creates a situation where anticipated sales (the dependent variable) help determine the final_price, generating simultaneity and thus reverse causality.

- **Measurement Error:** Although the dataset provides a recorded final_price, the value may not perfectly reflect the exact price customers encounter at the moment of purchase, especially when prices fluctuate rapidly. If a proxy (such as a competitor's price used to approximate Amazon's own pricing) were substituted, the proxy would likely embed noise, producing measurement error that biases the coefficient.

Q6. If the discount each week in our dataset is randomized by Amazon each week, will Equation 1 give the causal effect of price on sales? Give your reasoning. (**6pts**; 200 words)

If Amazon were to randomize the discount each week and therefore the final_price then Equation 1 would indeed identify the causal effect of price on sales. Randomizing the treatment ensures that the variation in final_price is unrelated to any other observed or unobserved determinants of demand. Because the discount would be assigned randomly across similar weeks or products, the resulting final_price becomes exogenous rather than influenced by Amazon's strategic decisions or market conditions.

In essence, randomized controlled trials (RCTs), including A/B price experiments, are viewed as the strongest approach for establishing causal inference. Randomization guarantees that the treatment (final_price) is statistically independent of factors such as product quality, demand shocks, or inventory fluctuations. This eliminates selection bias and balances all confounding variables both measurable and unmeasurable between the treated and control groups.

Mathematically, this setup allows the observed difference in sales between randomly assigned price levels to represent the unbiased Average Treatment Effect (ATE). As a result, the OLS coefficient on final_price would capture the true price elasticity of demand. With the manipulation completely random, endogeneity concerns such as omitted variable bias or reverse causality vanish, enabling Equation 1 to yield a valid causal estimate.

> ***Q7.*** From the below regression designed by another data scientist, discuss whether customers always prefer larger screens (i.e., everything else being equal, a larger screen always leads to higher sales)? (**4pts**; 150 words)

Yes, as per the outcomes of the regression analysis, the customers generally prefer larger screens, which means that all else being equal moving to a larger screen size leads to a statistically significant increases in sales.

Larger screens tend to create more sales, but this does not imply customers always prefer them. The regression includes factors (screensize) as an explanatory variable, while holding constant (controlling for) final_price, marketing_expense, and fixed effects for brand, technology, resolution and support_HDR. The phrase 'everything else being equal' is associated by this multivariate regression specification.

Comparing to the omitted baseline screen size category (which is smallest category not explicitly listed, approximately upto 29 inches or between 30 to 39 inches, the coefficients for th larger screen sizes are all positive and highly statistically significant too.

Sales Increased per units:

40-49 inches -> 48.601 *** units

50-59 inches -> 143.811 *** units

60+ inches -> 81.631 *** units

The result above demonstrates a preference for larger TV screens, as moving to any of the larger categories significantly boosts sales after controlling for technical features and price. Suprisingly, the 50-59 inch screen size provides the largest boost to sales (143.811 ***) among the categories listed, indicating that this specific size is the most preferred segment, even more so that the largest 60+ inch category.

```r
#given from the assignment brief
modelsummary(
  feols(
    fml = sales ~ final_price +
      factor (screensize) +
      marketing_expense |
      brand +
        technology +
        resolution +
        support_HDR,
    data = data_full
  ),
  stars = T,
  gof_map =c("nobs", "r.squared")
)
```

	(1)
final_price	0.672***
	(0.012)
factor(screensize)40-49	48.601***
	(7.227)
factor(screensize)50-59	143.811***
	(7.392)
factor(screensize)60+	81.631***
	(7.970)
marketing_expense	0.101***
	(0.002)
Num.Obs.	2080
R2	0.821
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

3. Instrumental Variables (20pts)

Q8. One way to obtain causal effects of price on sales from secondary data is to use the instrumental variable method. (**12pts**)

- List two variables you you would collect as instrumental variables for `final_price`

Make sure that an instrumental variable (Z) must be satisfied in three criteria relative to the endogeneous variable (X which is final_price) and the outcome variable (Y that is sales)

Criteria:

1. Exogeneity: Z must be uncorrelated with the error term (ϵ), meaning cov(Z,ϵ) equals 0. This means Z must be beyond individual or firm control and uncorrelated with unobserved confounding factors. Potential sources of IVs include environmental disasters and government policies.
2. Relevance: Z must be correlated with X, meaning is not cov(Z,X) is not 0.
3. Exclusion Restriction: Z must affect the outcome Y only through X.

Discussing 2 variables that satisfy the requirements for instrumenting final_price (X) in the sales regression (Y):

3.1 1. Unexpected Manufacturing Cost Shocks (External Variable)

Exogeneity: Sudden increases in manufacturing costs (e.g component shortages, global shipping spikes) represent external shocks unrelated to Amazon's internal pricing strategy or product characteristics. These shocks compel Amazon to adjust its recommended retail price (RRP) or modify discount levels to maintain margins, thereby creating exogenous variation in final_price.

Exclusion Restriction: These cost shocks should affect the quantity of TVs sold only by altering the final price shown to consumers. They do not directly influence demand drivers such as customer preferences or product quality. Thus, their only pathway to sales is through final_price, satisfying the exclusion restriction.

3.2 2. Competitor Price Index Thresholds (External Variable)

Exogeneity: Sharp, unanticipated changes in a major competitor's price index unrelated to Amazon's product quality or demand are typically driven by external market or policy factors. Since Amazon's algorithmic pricing often reacts to competitor movements, such exogenous competitor adjustments generate independent fluctuations in Amazon's final_price, fulfilling exogeneity.

Exclusion Restriction: Competitor price index shifts should influence Amazon's sales only through the price adjustment Amazon makes in response, not through any direct effect on consumer behaviour. Thus, the instrument affects sales solely through final_price, meeting the exclusion restriction.

- Can one use the VAT tax rate of TV products as an instrument variable for `final_price`? (**4pts**; 100 words)

Yes, the VAT tax rate is a strong candidate for an instrumental variable for final_price. Because VAT is set by government policy and cannot be influenced by Amazon's pricing algorithms or local demand conditions, it satisfies the exogeneity requirement. Changes in VAT directly affect the final price consumers pay, ensuring relevance. Additionally, VAT affects sales only by altering the displayed price, not by modifying the inherent quality of the product or consumer preferences. Therefore, the VAT rate also meets the exclusion restriction, making it an appropriate instrumental variable.

Q9. Assume you have identified one instrument variable `cost_shifter` in `data_full`. In the code blocks below, write down the two regressions you would need to run in order to estimate the causal effects of `final_price` on `sales`, including `marketing_expense` as the only control variable (**8pts**)

- Correct first stage codes and explanation of the code (**3pts**)
- Correct second stage codes and explanation of the codes (**3pts**)

```
# show the estimation code below and describe the steps

### Stage 1: First-stage regression
ols_stage1 <- feols(
  final_price ~ cost_shifter + marketing_expense,
  data = data_full
)

# Add predicted final_price to the dataset
data_full <- data_full %>%
  mutate(predicted_TV = predict(ols_stage1, data_full))

### Stage 2: Second-stage regression
ols_stage2 <- feols(
  sales ~ predicted_TV + marketing_expense,
  data = data_full
)

#Two-stage least squares (2SLS) method was employed to overcome the
endogeneity issues occuring in OLS regression
#first stage uses exogeneous instrument, cost_shifter to isolate the portion
of the final_price variation that is independent of unobserved confounder
which would result in predicted var as predicted_TV
#second stage estimates causal relationship by the regression on sales on
exogeneous component (predicted_TV)
#coeff for predicted_TV in second stage regression is -0.734***
#coeff represents unbaised causal relation of a unit change in final_price on
sales, holding marketing_expense constant, shwoing significant result-
statistically.

# do not modify the code below; this is to print out the results

modelsummary(list(
  "First Stage" = ols_stage1,
  "Second Stage" = ols_stage2
),
stars = T, gof_map = c('nobs','r.squared'))
```

	First Stage	Second Stage
(Intercept)	256.793***	1665.622***

	First Stage	Second Stage
	(40.790)	(63.067)
cost_shifter	1.099***	
	(0.066)	
marketing_expense	0.000	0.093***
	(0.003)	(0.003)
predicted_TV		-0.734***
		(0.068)
Num.Obs.	2080	2080
R2	0.118	0.306
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		

- Based on the results of the two regressions, discuss the causal effect of `final_price` on `sales` (**2pts**)

The IV estimate offers an essential correction to the original naïve analysis. In Equation 1 (from Q4), the OLS results showed a positive and counter-intuitive coefficient of +0.618 for **final_price**. This upward bias arose from endogeneity, where price was effectively capturing omitted factors such as product quality or underlying demand conditions. By introducing the instrumental variable (**cost_shifter**), the 2SLS approach eliminates this selection bias and isolates truly exogenous variation in price. As a result, the IV estimate uncovers the genuine negative causal effect of –0.734. This stark contrast highlights how OLS estimates based on observational data can be misleading when attempting to draw causal conclusions.

Q10. Design the A/B/N testing (**20 pts**)

Step 1:

The **unit of randomization** determines the level at which the treatment (e.g., an AI feature) is assigned—such as the individual, household, store, or city level.

Rationale:
For an online feature like Virtual Try-On, the most practical randomization unit is the individual customer or user ID. Using individual-level randomization maximizes the effective sample size, boosts the likelihood of achieving balance across groups, and reduces the risk of systematic differences in covariates before the experiment starts.

Disadvantages:
Fine-grained randomization introduces risks of **spillover** and **crossover** effects:

- **Spillover Effects (Violation of No Interference):**
  When randomization occurs at the individual level, treated users may influence friends or family—who may be in the control group—by sharing their positive or

negative experiences. This contaminates the control group and violates the SUTVA no-interference assumption.

- **Crossover Effects (Violation of Consistency):**
  Crossover occurs if a single participant is unintentionally exposed to both treatment and control conditions (e.g., Treatment A and Treatment B). This often happens online if cookies reset or the customer uses several devices, introducing hidden variation and breaking the consistency assumption of SUTVA.

  To reduce these risks—especially spillovers—Amazon could randomize at the **household** level instead, though this increases cost and complexity. When spillover concerns are minimal, the **individual level** remains the preferred option because it maximizes statistical power and experimental efficiency.

Step 2:

```
#below is an example of a simple, reproducible randomisation rule using the last digit of a user ID to place customers into buckets:
data_users <-  data_users%>%
  mutate(
    bucket = ID %% 10, #user ID's last digit
    treatment_A = ifelse(bucket ==0,1,0), #10% of treatment A
    treatment_B = ifelse(bucket ==1,1,0), #10% of treatment B
    control = ifelse(bucket >= 2,1,0) #80% of control
  )
```

```
Error: object 'data_users' not found
```

```
# 10/10/80 split balances experimental risk with the need for statistical power, mainly because the rule is deterministtic that also guarantees that each user stays in their assigned grp throughout the study.
```

Step 3:

```
#to determine how many users are needed per grp, we translate the assumed or expected treatetment lift into a standardised effect size, suppose treatment A is expected to have increament in avg spending by 10, with user level spending having a SD of 100

#d = 10/100 =0.1

#given two-sample t test with 80% of power and significance of 5%

pacman::p_load(pwr)
```

```
Installing package into 'C:/Users/VEDANT M BHATIA/AppData/Local/R/win-library/4.5'
(as 'lib' is unspecified)
```

```
Warning: unable to access index for repository
http://www.stats.ox.ac.uk/pub/RWin/bin/windows/contrib/4.5:
  cannot open URL
'http://www.stats.ox.ac.uk/pub/RWin/bin/windows/contrib/4.5/PACKAGES'

package 'pwr' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
    C:\Users\VEDANT M
BHATIA\AppData\Local\Temp\RtmpKGYhw7\downloaded_packages


pwr installed

Warning: package 'pwr' was built under R version 4.5.2

effect_size <- 0.1
alpha_value <- 0.05 # value given
power <- 0.8 #valeu given

sample_size_A <- pwr.t.test(
  d =effect_size,
  sig.level = alpha_value,
  power = power,
  type = "two.sample",
  alternative = "two.sided"

)
sample_size_A


     Two-sample t test power calculation

              n = 1570.733
              d = 0.1
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group

#generates minimum number of users per group needed to reliably detect
expected lift. similar calculation with a smaller lift ( eg 5 for treatmnt B)
requires more users significantly.
```

Step 4:

Two types of data must be collected, each serving a distinct purpose:

1. Pre-treatment data (for randomisation checks)

- Demographics (age, gender, location, if available)
- Historical purchasing behaviour
- Past browsing patterns
- Device type, Prime membership status

These variables help confirm that randomisation successfully created comparable groups before the experiment began.

2. In-experiment outcome data (for measuring the treatment effect)

- Total spending during the experiment
- Number of orders placed
- Conversion rate
- Return rate
- Engagement with the Virtual Try-On feature (e.g., usage frequency, time spent)

Collecting these outcomes at the user level aligns with user-level randomisation and ensures that causal effects can be estimated cleanly and accurately.

Step 5:

Drawing conclusions from the statistical analysis:

```
#Randomising check usign pre-treatment covariates
t.test(baseline_spend ~ treatment_A,
       data = data_users %>% filter(control ==1))

Error in eval(m$data, parent.frame()): object 'data_users' not found

t.test(baseline_spend ~ treatment_B,
       data = data_users %>% filter(control ==1))

Error in eval(m$data, parent.frame()): object 'data_users' not found

#compare group means
data_users %>%
  group_by (control, treatment_A, treatment_B)%>%
  summarise(mean_spend = mean(spend), .groups = "drop")

Error: object 'data_users' not found

#estimating treatments A and B effects via regression
pacman::p_load(dplyr, fixest, modelsummary)
```

```r
data_users <- data_users %>%
  mutate(
    treatment_factoring = case_when(
      treatment_A == 1 ~ "A",
      treatment_B == 1 ~ "B",
      TRUE ~ "control"
    ),
    treatment_factoring = factor(treatment_factoring,
                                 levels = c("control", "A", "B"))
  )
```

```
Error: object 'data_users' not found
```

```r
model_abntest <- feols(
  spend ~ treatment_factoring,
  data = data_users
)
```

```
Error in feols(spend ~ treatment_factoring, data = data_users): Argument
'data' could not be evaluated.
PROBLEM: object 'data_users' not found.
```

```r
modelsummary(model_abntest, stars = TRUE)
```

```
Error: object 'model_abntest' not found
```

```r
#coeff on treatment factor of A and factor B give estimate change in spendign
relative to control group
```

Q11. Finally, Tom would like to study the causal effect of Amazon rating on product sales. For instance, what is the causal effect of a 4.5-star rating on sales compared to a 4-star rating. Propose **one** natural experiment method to study this causal question. (**12pts**)

A strong natural experiment for this context is a **Regression Discontinuity Design (RDD)**. Amazon upgrades the visual star rating (e.g., to 4.5 stars) only when a product's underlying continuous rating surpasses a fixed cutoff (such as 4.50). Products positioned just above or below this threshold are nearly identical, and sellers cannot precisely adjust their true average rating to manipulate placement relative to the cutoff. As a result, assignment to the higher displayed rating is effectively random in a narrow region around the threshold. Any discontinuous change in sales at this point can therefore be interpreted as the causal effect of receiving the enhanced visible rating.

To implement RDD, product-level data are needed near the cutoff, including:

- **Running variable:** the exact numerical rating score (e.g., 4.498).
- **Treatment indicator:** whether the product is shown with the higher visual rating (1 = 4.5 stars, 0 = 4.0 stars).

- **Outcome variable:** sales (units or revenue).
- **Covariates:** features such as price, brand, or screen size, used to confirm smooth evolution around the threshold.

```
The dataset must span a reasonable bandwidth around the cutoff to ensure
comparability. RDD proceeds by limiting the sample to a narrow window (e.g.,
4.4–4.6) so that products differ only in treatment status. Balance tests
should verify that covariates do not shift discontinuously at the threshold;
if they are smooth, any jump in sales identifies the causal effect of the
higher star rating.
```

We then estimate a **local linear regression:**

1. Statistical Analysis Steps:

Select Bandwidth:
Choose a narrow interval (bandwidth) around the cutoff (e.g., rating scores from 4.4 to 4.6) to ensure that only products that are genuinely comparable are included in the analysis.

Randomisation Check:
Confirm that product characteristics (covariates) remain smooth and balanced on either side of the threshold within the selected bandwidth. This ensures there are no systematic differences between products just above and just below the cutoff aside from the visual rating itself.

Local Regression:
Estimate a linear regression model restricted to the chosen bandwidth to identify the discontinuity in sales at the cutoff point.

2. Regression Specification:

The primary RDD model, using a local linear framework, can be written as:

Salesi = β0 + β1Rating_Visuali + β2Scorei + ϵi

where:

- $\textbf{Sales}_i$ = the dependent variable (sales volume).
- $\textbf{Rating\_Visual}_i$ = treatment indicator (1 if the product shows the higher visual rating, 0 if not).
- $\textbf{Score}_i$ = the running variable (actual average rating score).

The parameter $\boldsymbol{\beta_0}$ represents the Local Average Treatment Effect (LATE), which captures the size of the sales jump at the cutoff—i.e., the causal impact of receiving the higher displayed star rating compared to the lower one.